

Other industrial devices that have real-time control functions like a robot are more safety orientated. The controller part probably consumes a small percentage of the energy compared to the motor drives. Therefore controller power saving is less desirable.

How do you deal with component end-of-life issues? How can an engineer better prepare himself or his company to quickly deal with this situation?

In researching the best way to manage component end-of-life issues, Emerson has found that there are two separate activities. First, minimise the issue by matching the component lifecycle to the product lifecycle. We do that as a design requirement, where products get a defined expected lifecycle and the design engineers do their best to select components with lifecycles that match our product's lifecycle.

Second, when a component reaches its end-of-life before the product, we have to react. Emerson has a dedicated staff to monitor component changes, including discontinuance. The change information is then sent to a dedicated component engineering team, which evaluates replacements and then provides information to the design engineering team. Ideally, replacement components are available that offer little change to the product, but that is not always true.

In summary, the easiest way to reduce component end-of-life issues is to design longevity into the product. Design choices cannot guarantee elimination of the issue but can minimise the disruption.

How soon can we expect Android to be stable enough for the industrial automation market and the embedded space?

Android is already being used in some of the more consumer-facing embedded applications. It is well suited for those applications that require very intuitive operation. For example, home medical equipment, touch-controlled signage, gaming and even some aircraft seatback-entertainment systems.

There are industrial applications that need similar, easy-to-use controls, but many of these are also running real-time control algorithms that don't sit so well with Android. From the hardware perspective, we have seen a major change to multi-core processors and over time we will see more use of virtualisation in the embedded space with multiple operating systems, and this is likely to increase the use of Android as the user interface in industrial automation applications.

In industrial automation projects, engineers require different application-specific interfaces because of the greater level of integration. What interfaces do you recommend?

Splitting this into two halves, there are the upstream interfaces that enable factory-wide connectivity and ancillary

communication, and then the downstream interfaces that are very application-specific.

The dominant upstream interface is Ethernet for distributed connection to associated equipment as well as to factory-wide planning and financial systems. Very generally, there are two types of Ethernet connectivity: commercial Ethernet, as found in equipment such as laptops, and industrial Ethernet, which has additional functionality for time stamping and determinacy that makes it better suited for real-time applications. In addition to Ethernet, many applications require standard interfaces such as USB and MicroSD to configure and store user programs.

The local downstream interfaces are more application-specific. RS422/485/232 serial interfaces are still widely used by legacy embedded sensors and equipment. Motor control is extremely relevant and so local stepper motor interfaces can be provided along with AC and DC drive controllers. CANbus is quite popular, particularly if the application involves vehicle manufacture and testing. In addition, direct digital and analogue input/output are often used for both measurement and control.

Is there any technology from other domains that could benefit industrial automation as well?

One specific technology that is still to be widely deployed in the industrial space is remote management over Ethernet. Despite many products having this capability and it being widely used in other embedded markets like telecom, its financial and operational benefits are still to be realised in industrial control.

Emerson recently launched a plug-in media-processing accelerator. What drives the need for 'network media processing' here?

There has been tremendous growth in data and video over mobile networks. As networks transition to an all-IP environment, service operators are finding a need for new advanced flow-management devices such as session border controllers, QoS analytic engines and intelligent flow optimisers.

The ability to have media processing, such as voice transcoding within the box, helps to simplify the communications flow for an operator, hence providing a competitive advantage for the equipment vendor. Unfortunately, voice stream processing in real time at high channel counts is a strenuous task. So adding this function can impose a significant reduction on the processing power available to the main service, leading to a reduction in capacity.

A more elegant solution to the problem is to use a plug-in media processing accelerator to offload both audio and video processing from the server host. In this case, using a plug-in media-processing accelerator offers a true upgrade path. ●

(The complete interview can be read at electronicsforu.com)